Stage 4b: Partial Rebound and Model Ensembles

Question 1. What does the rebound do, and why does it only matter around January 4?

Step 1. Reproduction number after reopening

Given R_t = 1.260, s_now = 0.910 and s = 0.808 on January 7:

```
s/s_now = 0.808/0.910 = 0.8879121 (approximately 0.888).
```

```
R_after = 1.260 x rho x (0.808/0.910).
```

rho	Substitution	R_after
1.0	1.260 x 1.0 x 0.887912	1.119
0.9	1.260 x 0.9 x 0.887912	1.007
0.8	1.260 x 0.8 x 0.887912	0.895

rho = 1.0 and rho = 0.9 give R_after greater than one and therefore permit renewed growth. rho = 0.8 gives R_after less than one, so infections decline under the stated assumptions.

Step 2. Why rho cancels in February

Once all 21 infection days behind the measured three-week growth fall after the break, the average multiplier is rho. Therefore:

```
f_avg = rho; beta_school = beta_now/rho;
```

```
beta_future = beta_school x rho = (beta_now/rho) x rho = beta_now.
```

The multiplier is present in both the estimate of term-time transmission and the future projection, so it cancels. It primarily affects forecasts that cross January 4, rather than those made entirely after the three-week estimation window has moved beyond the break.

Step 3. Mid-January forecast bump

Scenario	Predicted admissions/week
rho = 1.0	about 46,000
rho = 0.9	about 38,000
Observed	about 20,000

The bump decreases by approximately 46,000 - 38,000 = 8,000 admissions, or (8,000/46,000) x 100 = 17.39%. It is not eliminated: early-January forecasts still extrapolate the high growth measured just before the winter break.

Question 2. Who wins and who loses?

Step 1. Percentage change, Stage 4 to Stage 4b

```
Percentage change = (new - old)/old x 100. Negative values indicate improved (lower) WIS.
```

```
2023-24: (0.641 - 0.783)/0.783 x 100 = -18.14%.

2024-25: (0.792 - 0.747)/0.747 x 100 = +6.02%.

2025-26: (0.656 - 0.812)/0.812 x 100 = -19.21%.
```

Season	Stage 4	Stage 4b	Change
2023-24	0.783	0.641	-18.14%
2024-25	0.747	0.792	+6.02%
2025-26	0.812	0.656	-19.21%

Step 2. Why 2024-25 worsens

The real 2024-25 peak was in February 2025. rho = 0.9 reduces post-break transmission to 90% of the pre-break level in forecasts crossing reopening, which suppresses the predicted rebound and contributes to underprediction of the February peak.

Step 3. Fair-test rule

The parameter was selected on 2023-24 and 2024-25, with 2025-26 held out. For kappa = 0.3 and holiday factor 0.8, the tuning means are 0.765 (rho = 1.0), 0.716 (rho = 0.9), and 0.793 (rho = 0.8). Thus rho = 0.9 was selected. Switching to rho = 0.8 because it scored 0.606 on the held-out season would use the test outcome to retune the model and invalidate the independent test.

Step 4. Decline bias

Bias = forecast median - observed. Positive bias means predictions are too high. The decline bias falls from +92 to +34, a reduction of 92 - 34 = 58 admissions, or 58/92 x 100 = 63.04%. The smaller predicted rebound reduces excessive post-break admissions forecasts.

Question 3. Why did the ensembles not help?

Step 1. Compare tuning means

Model	Tuning mean
A (kappa 0.3, 0.4)	0.761
B (rho 0.8, 0.9, 1.0)	0.726
C	0.769
D	0.769
Single model	0.716

The single model has the lowest tuning mean, 0.716. Ensemble B is closest at 0.726.

Step 2. Why kappa = 0.4 hurts

For holiday factor 0.8 and rho = 0.9 in 2024-25, kappa = 0.4 scores 1.036 versus 0.792 for kappa = 0.3. That season was 2.31 times a typical season. Because s_now = 1 - kappa(H/H_bar), larger kappa infers faster susceptible depletion for the

same cumulative admissions, making the simulated epidemic turn too early in an unusually large season.

Step 3. Interval coverage

The national 95% intervals contained every observed outcome (100% coverage). This suggests overly wide intervals, although coverage alone does not prove the optimal width. Mixing different forecast distributions tends to widen the intervals further; with coverage already above the nominal 95%, this can increase WIS without meaningful coverage benefit.

Step 4. When an ensemble might be useful

When the appropriate kappa for a new season is uncertain, an ensemble spreads forecasts across plausible susceptible-pool assumptions and may improve robustness. It remains worth testing live as a non-designated model, while preserving the better-scoring single model as the selected submission.